

Instructions to Candidates:

- 1) Attempt any five questions.
- 1) Illustrate the answers with neat sketches, diagram etc. wherever necessary.
- 2) Necessary data is given in the respective questions. If such data is not given it means its knowledge is a part of examination.
- 3) If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.

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- Q 1 A Define communication, Draw and explain basic block diagram of communication system. (6M)
- B Define modulation; Explain the need of modulation in detail. (6M)
- 2 A For an AM wave with a peak unmodulated carrier voltage $V_c = 10V_p$, a load resistance $R_L = 10\Omega$, and modulation index = 1, determine (8M)
- a. Powers of the carrier and the upper and lower sidebands
 - b. Total sideband power.
 - c. Total power of the modulated wave
 - d. Draw the power spectrum.
- Repeat steps (a) through (d) for a modulation index $m = 0.5$.
- B Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent, and (b) 50 percent. (4M)
- 3 A Draw the block diagram for an Armstrong indirect FM transmitter and describe its operation. (8M)
- B Compare narrowband and wideband FM. (4M)
- 4 A With the help of neat block diagram explain functioning of a super heterodyne receiver list out significance. (6M)
- B Explain the performance characteristics of receiver. (6M)
- 5 A Explain the linear diode (envelope) detector with detail circuit diagram and characteristics. (6M)
- B With neat circuit diagram and necessary equations, explain the phase difference discriminator ratio detector. (6M)
- 6 A An amplifier has a noise figure of 4 dB, a bandwidth of 500Hz and an input resistance of 500Ω . Calculate the input signal voltage needed to yield an output $SNR = 1$ when the amplifier is connected to a signal source of 50Ω at 290K. (4M)
- B Define noise. Explain the classification of noise in detail. (8M)

Paper End
